# REPORT 5 - NOSE CONE 720 MM LOADS FOR STRUCTURE

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**Weisshorn – 720 mm Nose Cone: Pressure, Forces and Heating**

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# 1 Results for the structure team

This report gives ST the aerodynamic and thermal inputs to size the two 720 mm noses. ST computes the joint and shell loads from them. Both noses are 720 mm long with $D = 180$ mm and a 4.5 mm tip radius; their geometry is in `2027_C_FD_NOSE-CONE-720MM_GEOMETRY`.

- **A** (baseline, 3U inside the nose): LV-Haack, 679.0 mm + 41 mm coupler.
- **B** (backup, 3U elsewhere): Von Kármán, 700.0 mm + 20 mm coupler.

The numbers come from eight Fluent runs (2D axisymmetric RANS, zero incidence) at $M = 0.95$, 1.20 and 1.47 (burnout), and from the Python flight model, calibrated on these runs, for the time histories and the hot day (30 °C, LVO masses). The nose is **sealed**: the air inside stays at the pad pressure $p_{pad} = 101.3$ kPa, raised by its heating,

$$p_{in} = p_{pad}\,\frac{T_{in}}{T_{pad}}, \qquad \Delta p = p_w - p_{in}, \tag{1}$$

between $T_{in} = T_{pad}$ and $T_{in}$ = mean temperature of the inner face of the shell (upper bound). $\Delta p > 0$ pushes the shell inwards. Axial forces are positive when they push the nose aft onto the joint (compression).

Table 1: Load cases. Pressures: tip with $p_{in}$ at $T_{pad}$ (largest inward load), shell minimum with heated air (largest outward load). Net pressure force on the joint: $\sum (p_w - p_{in})\, 2\pi r\, \mathrm{d}r$ over the outer surface, for $T_{in} = T_{pad}$ / heated air. $D_n$: aerodynamic axial force of the nose (model, 20 µm roughness). $N/\alpha$: normal force per degree of incidence.

| | A nominal | A hot day | B nominal | B hot day |
|---|---|---|---|---|
| **LC1, burnout** ($M_{max}$, $q_{max}$) | | | | |
| Time $t$ [s], Mach $M$ | 9.1, 1.47 | 9.1, 1.56 | 9.1, 1.47 | 9.1, 1.56 |
| Dynamic pressure $q$ [kPa] | 116.8 | 131.4 | 117.5 | 132.4 |
| $\Delta p$ at the tip [kPa] | 153 | 175 | 154 | 177 |
| Lowest $\Delta p$ on the shell [kPa] | -46 | -49 | -48 | -50 |
| Net pressure force on the joint [N] | -397 / -753 | -374 / -767 | -436 / -793 | -418 / -813 |
| Aerodynamic axial force $D_n$ [N] | 278 | 310 | 234 | 260 |
| Normal force $N/\alpha$ [N/°] | 104 | 117 | 104 | 118 |
| **LC2, transonic** ($M = 0.95$ on the ascent) | | | | |
| Time $t$ [s], $q$ [kPa] | 5.9, 57.6 | 5.8, 57.9 | 5.9, 57.3 | 5.8, 58.0 |
| $\Delta p$ at the tip [kPa] | 63 | 64 | 62 | 64 |
| Lowest $\Delta p$ on the shell [kPa] | -22 | -21 | -29 | -28 |
| Net pressure force on the joint [N] | -265 / -358 | -248 / -335 | -263 / -355 | -246 / -333 |
| Aerodynamic axial force $D_n$ [N] | 53 | 51 | 44 | 47 |
| Normal force $N/\alpha$ [N/°] | 51 | 51 | 51 | 52 |
| **LC3, apogee** ($q \approx 0$) | | | | |
| Time $t$ [s], $p_\infty$ [kPa] | 43.7, 31.3 | 44.2, 31.1 | 43.9, 30.8 | 44.5, 30.5 |
| $\Delta p$ on the whole shell [kPa] | -70 | -70 | -71 | -71 |
| Net pressure force on the joint [N] | -1783 | -1788 | -1794 | -1802 |
| Nose centre of pressure $x_{CP}$ from the tip [mm] | 295 | | 347 | |

Table 2: Heating. Fluent values: burnout, nominal, wall at 300 K. Maximum temperatures over the flight: Python model calibrated on Fluent (Section 5), CFRP shell 1.2 mm, Al 6082 tip insert 80 mm.

| | A nominal | A hot day | B nominal | B hot day | Limit |
|---|---|---|---|---|---|
| Heat flux at the tip $q_w$ [kW/m$^2$] (Fluent) | 172.5 | – | 173.4 | – | |
| $h$ at the tip / shoulder [W/(m$^2$K)] (Fluent) | 1888 / 546 | – | 1899 / 539 | – | |
| Stagnation temperature $T_0$ [°C] | 118 | 154 | 119 | 156 | |
| Al tip insert [°C] | 61 | 85 | 56 | 79 | |
| Shell outer face, 50 mm [°C] | 82 | 110 | 82 | 110 | 120 (DP490) |
| Shell inner face, 50 mm [°C] | 77 | 104 | 78 | 104 | 67 (Req_10) |
| Inside air, upper bound [°C] | 72 | 98 | 73 | 99 | 67 (Req_10) |

**What this means for the structure.**

1. **The sealed nose is the largest load.** At apogee the outside pressure has dropped to 31.3 kPa while the inside stays at the pad pressure: $\Delta p \approx -70$ kPa on the whole shell, which pulls the nose forward off the joint with about 1.8 kN, close to the 2000 N of Req_07. At burnout the inside overpressure already outweighs the drag: the net pressure force is -397 to -753 N (tension). Vent holes would bring $p_{in}$ to $p_\infty$ and remove this load.
2. **Inward pressure** is highest at the tip at burnout, 153 kPa (175 kPa hot day), on the spherical cap only. Beyond $x = 2$ mm it is at most 36 kPa for A and 17 kPa for B (43 and 22 kPa hot day), and below 10 kPa after 50 mm (Figure 4).
3. **Aerodynamic axial force** peaks at burnout: 278 N for A and 234 N for B (310 and 260 N hot day). Fluent with a smooth wall gives 282.6 and 237.6 N.
4. **Normal force** under incidence $\alpha$: $N = 2\,q\,A_{ref}\,\alpha$, acting at $x_{CP}$, distributed as $\mathrm{d}N/\mathrm{d}x = 2\,q\,\alpha\,\mathrm{d}S/\mathrm{d}x$ (Section 4). ST chooses the design incidence (Req_12).
5. **Temperatures.** The DP490 limit (120 °C) is not reached. The inner face of the shell and the inside air exceed Req_10 (67 °C) already on a nominal day, for both designs.
6. **A against B.** The heating is the same within 1 %. B has 15 % less drag at burnout, about half the inward pressure on the front of the shell (item 2), and a 20 mm coupler instead of 41 mm. The sealed-nose loads are the same for both.

The data behind every figure are delivered as CSV tables (Appendix B).

## 2 Designs and flight points

Table 3: The two 720 mm noses. $L$: Haack length to the sharp apex; the 4.5 mm spherical tip removes $\delta$, so the curved part is $L - \delta$ long. Apogee: Python flight model calibrated on Fluent (Section 4).

| | A (baseline) | B (backup) |
|---|---|---|
| Profile | LV-Haack ($C = 1/3$) | Von Kármán ($C = 0$) |
| 3U payload | inside the nose | not in the nose |
| $L$ / $\delta$ [mm] | 682.3 / 3.3 | 705.5 / 5.5 |
| Curved part $x_S$ + coupler $c$ [mm] | 679.0 + 41 | 700.0 + 20 |
| Fineness $L/D$ | 3.79 | 3.92 |
| Apogee, nominal / hot day [m] | 8887 / 9397 | 8982 / 9518 |

**Flight.** The trajectory is the one of the Python design tool (report 2), with the nose of each design. Two cases are given everywhere:

- **nominal**: ISA day (288.15 K on the pad), propellant mass from the total impulse;
- **hot day**: 303.15 K on the pad and the LVO masses. The rocket is lighter and faster, so $q$ and the heating are higher.

Along the trajectory,

$$q = \tfrac{1}{2}\,\rho_\infty V^2 = \tfrac{1}{2}\,\gamma\, p_\infty M^2, \qquad T_0 = T_\infty \left(1 + \tfrac{\gamma-1}{2}\,M^2\right). \tag{2}$$

The highest Mach number, dynamic pressure and stagnation temperature all occur at burnout (Figure 1).

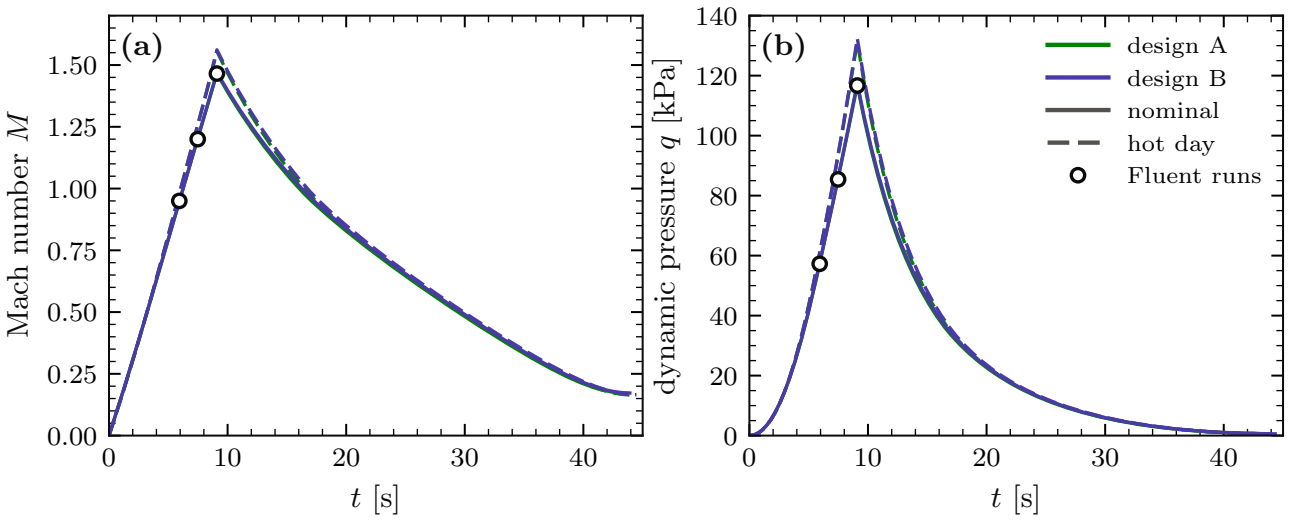



Figure 1: (a) Mach number and (b) dynamic pressure along the flight, both designs, nominal and hot day. Circles: the free stream of the Fluent runs.

**Fluent points.** The three Fluent runs use the free stream of design A's nominal trajectory at these instants (Table 4); design B is run at the same points, so that A and B differ by shape only (B's own burnout is at $M = 1.471$). Values at other times are interpolated in $M$ (Section 3).

Table 4: Free stream of the Fluent runs. Reynolds number per metre: $\rho_\infty V/\mu_\infty$.

| | Transonic | Supersonic | Burnout |
|---|---|---|---|
| Time $t$ [s] | 5.93 | 7.48 | 9.10 |
| Altitude $z$ [m] | 923 | 1476 | 2185 |
| Mach number $M$ | 0.950 | 1.200 | 1.465 |
| Static pressure $p_\infty$ [Pa] | 90714 | 84801 | 77685 |
| Static temperature $T_\infty$ [K] | 282.15 | 278.55 | 273.95 |
| Dynamic pressure $q$ [kPa] | 57.3 | 85.4 | 116.7 |
| Reynolds number per metre [$10^6$/m] | 20.3 | 24.4 | 27.9 |

**Load cases.** Three instants bound the aerodynamic loads of the sealed nose:

- **LC1, burnout**: highest $M$, $q$, inward pressure at the tip, axial force, normal force per degree and heating;
- **LC2, transonic**, $M = 0.95$ on the ascent: the shock system changes, lower $q$;
- **LC3, apogee**: $q \approx 0$ and the lowest outside pressure, so the largest outward (burst) pressure on the sealed nose.

Recovery (Req_07) and the inertial loads at the joint are not aerodynamic and are left to ST.

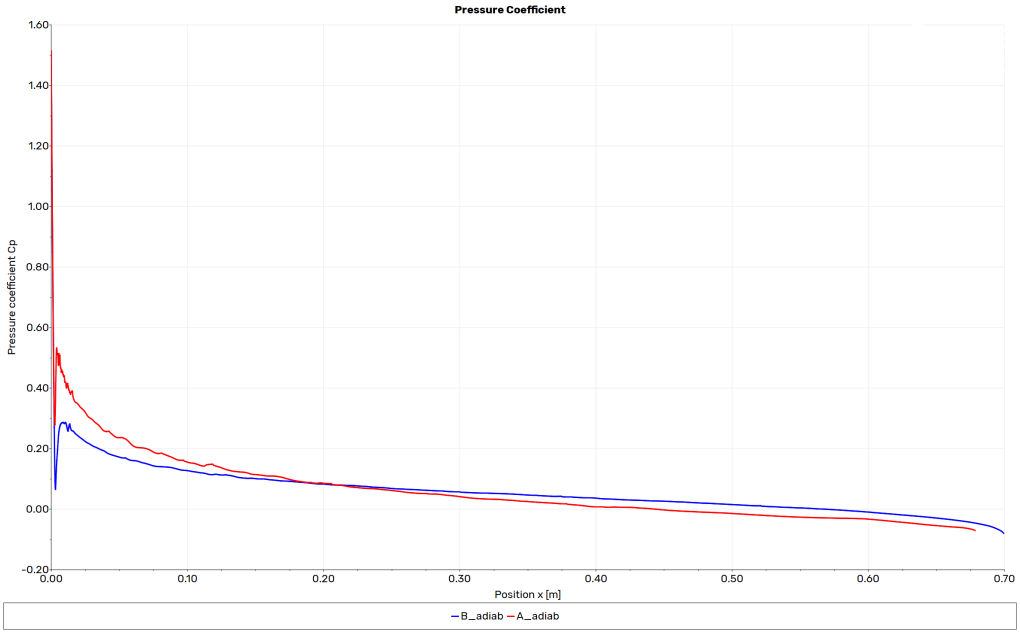
# 3 Pressure on the shell

## 3.1 Fluent wall pressure

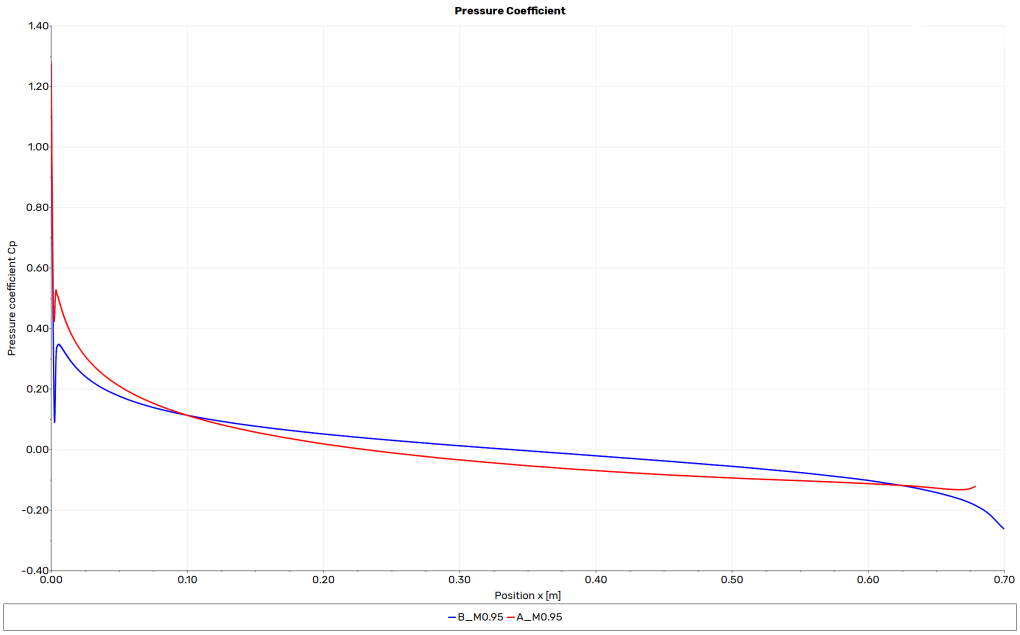
Fluent gives the wall pressure as a pressure coefficient, which depends on the Mach number but hardly on the altitude:

$$C_p(x) = \frac{p_w(x) - p_\infty}{q}. \tag{3}$$

At the stagnation point $C_p = (p_0 - p_\infty)/q$: 1.51 at burnout and 1.28 at $M = 0.95$ (Figure 2). The pressure then drops quickly over the spherical tip and the first centimetres, where A, with its fuller LV-Haack front, keeps about twice the pressure of B. Further aft both noses are close to $p_\infty$, and below it towards the shoulder, where the flow accelerates onto the cylinder: lowest $C_p$ -0.07 (A) and -0.08 (B) at burnout, -0.13 and -0.26 at $M = 0.95$.

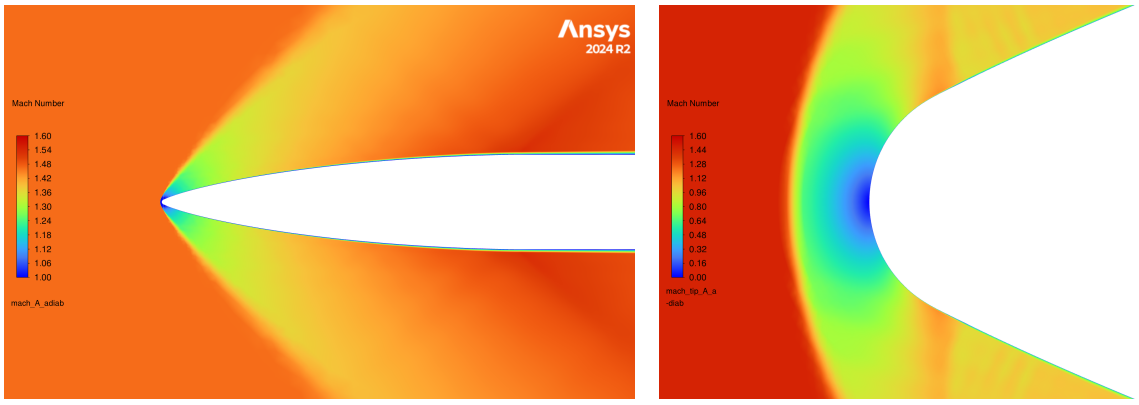



(a) Burnout, $M = 1.465$.




(b) Transonic, $M = 0.95$.

Figure 2: Wall pressure coefficient along `wall_nose`, A (red) and B (blue), Fluent XY plots. The $M = 1.20$ run lies between the two and is in the data tables.




(a) Nose of A. (b) Tip of A.

Figure 3: Mach number at burnout (Fluent). The bow shock stands just ahead of the spherical tip, with a small subsonic region behind it where the stagnation pressure acts.

### 3.2 From the Fluent runs to the whole flight

At any time $t$ of the trajectory, $C_p$ is interpolated linearly in $M$ between the three runs, and

$$p_w(x,t) = p_\infty(t) + C_p\big(x, M(t)\big)\, q(t), \qquad \Delta p(x,t) = p_w(x,t) - p_{in}(t), \tag{4}$$

with $p_{in}$ of Equation (1). Outside the Fluent range $C_p$ is held: below $M = 0.95$ (low $q$, so small loads) and above $M = 1.465$, which only the hot day reaches ($M = 1.56$). There the stagnation pressure comes out -1.7 % (A) and -1.9 % (B) below the Rayleigh pitot value; on the nominal trajectory the difference is -0.2 % and -0.4 %.

Between $x = 3$ and 15 mm Fluent's $C_p$ has a small sawtooth where the 0.2 mm tip cells of the mesh grow to the 1 mm wall cells. It is smoothed with a 2 mm moving average in the figures and tables below; the integrated forces are not affected.

### 3.3 Net pressure on the sealed nose

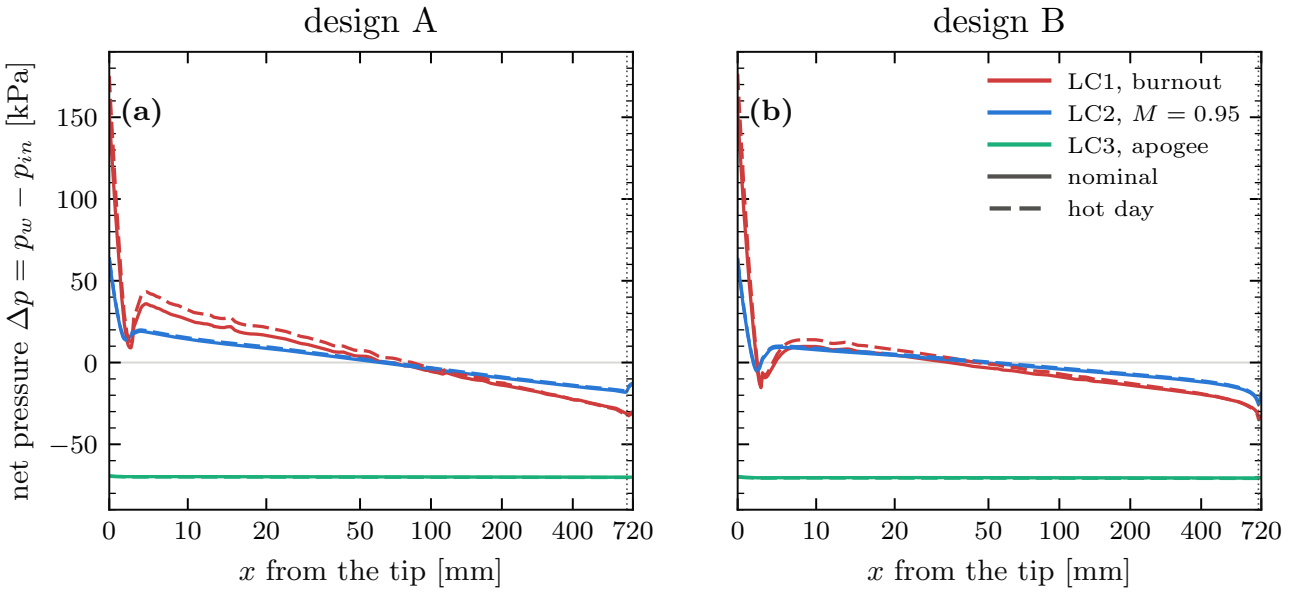



Figure 4: Net pressure along the nose at the three load cases, (a) A and (b) B. LC1 and LC2 with $p_{in}$ at $T_{pad}$; LC3 with both bounds of $p_{in}$ (thick: $T_{pad}$, thin: heated air). $x$ on a logarithmic scale above 20 mm; dotted: shoulder.

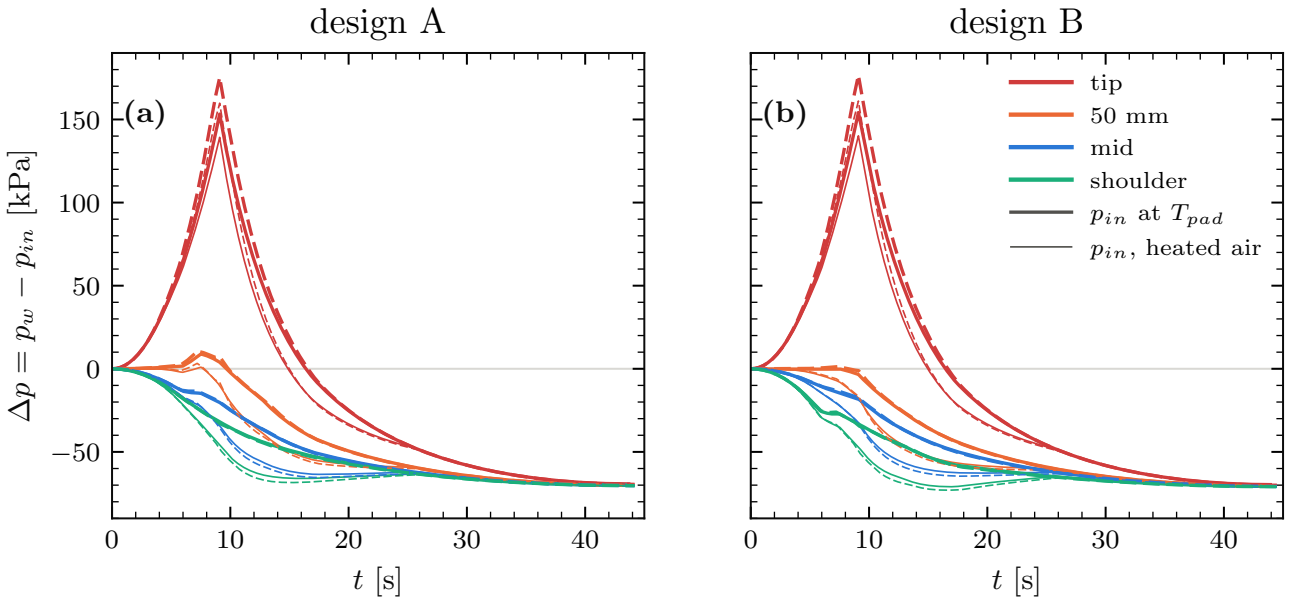



Figure 5: Net pressure along the flight at four stations, (a) A and (b) B. Thick: $p_{in}$ at $T_{pad}$; thin: heated inside air.

Table 5: Extremes of the net pressure over the flight at the four stations [kPa], with the time [s] in brackets. Inward: $p_{in}$ at $T_{pad}$; outward: heated inside air.

| | | A nominal | A hot day | B nominal | B hot day |
|---|---|---|---|---|---|
| Tip | inward | 153 (9.1) | 175 (9.1) | 154 (9.1) | 177 (9.1) |
| | outward | -69 (43.7) | -70 (44.2) | -70 (43.9) | -70 (44.5) |
| 50 mm | inward | 9 (7.5) | 11 (7.2) | 0 (7.5) | 2 (7.2) |
| | outward | -70 (43.7) | -70 (44.2) | -70 (43.9) | -71 (44.5) |
| Mid-length | outward | -70 (43.7) | -70 (44.2) | -71 (43.9) | -71 (44.5) |
| Shoulder | outward | -70 (43.7) | -70 (44.2) | -71 (17.0) | -73 (16.6) |

- **Inward** loads exist only on the front of the nose: the tip cap at burnout and, for A, the first centimetres (9 kPa at 50 mm, around $M = 1.2$). From mid-length to the shoulder the net pressure is never inward.
- **Outward** loads grow along the whole climb as $p_\infty$ falls, and reach about 70 kPa on the whole shell at apogee. Heating of the inside air adds a few kPa during the coast (thin lines in Figure 5); at the shoulder of B this gives the largest outward value, -73 kPa at $t = 16.6$ s on the hot day.
- Over the shell the two designs differ only at the front; at apogee they are identical, because there $\Delta p = p_\infty - p_{in}$ does not depend on the shape.

The full distributions $p_w(x)$ and $\Delta p(x)$ at LC1–LC3, and $\Delta p(t)$ at the four stations, are in the data tables (Appendix B).

# 4 Forces on the nose

## 4.1 Axial force from Fluent

The axial force is integrated over the Fluent wall faces, with $r$ the local radius:

$$F_p = \sum (p_w - p_\infty)\, 2\pi r \,\mathrm{d}r, \qquad F_v = \sum \tau_w \, 2\pi r \,\mathrm{d}x, \qquad C_D = \frac{F_p + F_v}{q\, A_{ref}}, \quad A_{ref} = \pi R^2 = 0.025\,447\ \mathrm{m}^2. \tag{5}$$

The sums reproduce Fluent's own force report to 0.1 N. $F_p$ is the pressure drag relative to $p_\infty$; the net pressure force on the joint of the sealed nose (Table 1) also contains $(p_\infty - p_{in})\, A_{ref}$.

Table 6: Axial force at the three Fluent points [N] (smooth wall), and $C_D$ of `wall_nose` (pressure + viscous). The coupler carries friction only.

| | A | | | B | | |
|---|---|---|---|---|---|---|
| $M$ | 0.95 | 1.20 | 1.465 | 0.95 | 1.20 | 1.465 |
| Pressure $F_p$, nose | 4.6 | 149.2 | 204.7 | 6.0 | 125.2 | 165.2 |
| Friction $F_v$, nose | 42.0 | 57.1 | 72.4 | 40.5 | 54.9 | 69.8 |
| Friction, coupler | 2.9 | 4.4 | 5.4 | 1.6 | 2.2 | 2.7 |
| **Total** | **49.6** | **210.6** | **282.6** | **48.2** | **182.3** | **237.6** |
| $C_D$ nose | 0.0320 | 0.0949 | 0.0933 | 0.0320 | 0.0828 | 0.0791 |
| B against A | | | | +0 % | -13 % | -15 % |

Below $M = 1$ the two noses have the same drag, about 90 % of it friction. Above $M = 1$ the wave drag dominates, and B's thinner front gives it 13 to 15 % less.

## 4.2 Drag model of the flight

The flight model (report 2) adds a wave drag from Euler simulations and a flat-plate friction,

$$C_{D,n}(M) = C_{D,w}(M) + C_f\,\frac{S_{wet}}{A_{ref}}, \qquad C_{D,w}(M) = C_{D,w}(1.1)\,s^2(3-2s), \quad s = \frac{M - M_{on}}{1 - M_{on}} \in [0,1] \tag{6}$$

below $M = 1.1$. Report 2 used $M_{on} = 0.8$, which gave 2.6 times Fluent's drag at $M = 0.95$ (0.0826 against 0.0320). With $M_{on} = 0.94$ the model now matches Fluent at all three points (Figure 6):

| Fluent / model $-1$ | $M = 0.95$ | 1.20 | 1.465 |
|---|---|---|---|
| A | -1.7 % | +1.7 % | +1.7 % |
| B | +4.2 % | +1.8 % | +1.9 % |

The change raises the apogee by 7 m (A) and 1 m (B). The rise between $M = 0.95$ and 1.1 is not resolved by Fluent; the model puts it between 0.94 and 1.0.

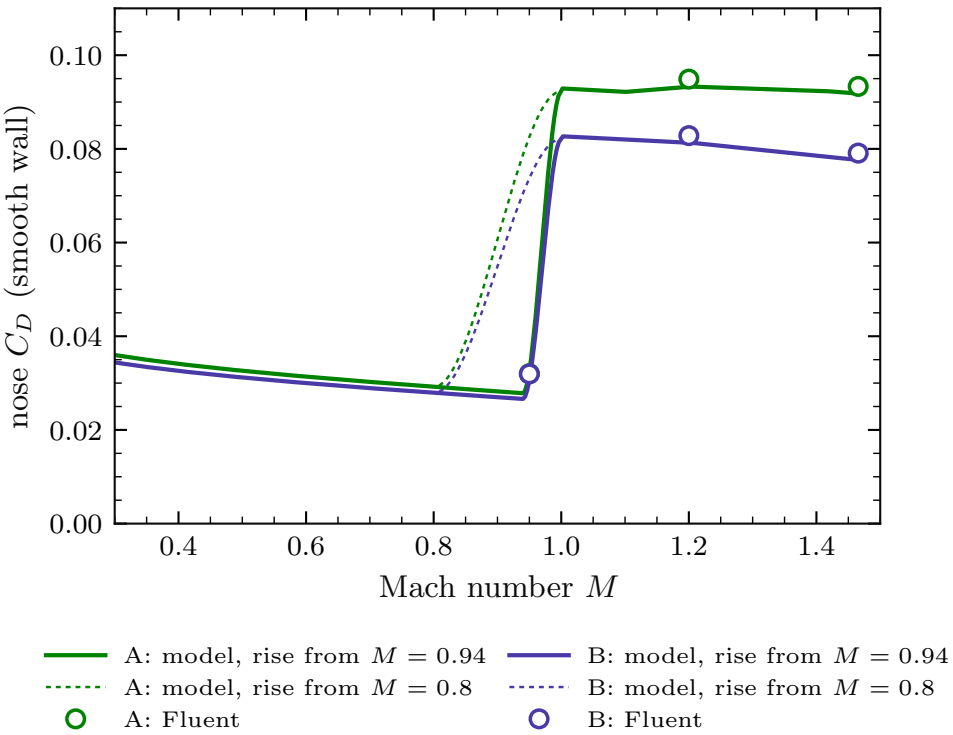



Figure 6: Nose $C_D$ against Mach along the ascent: flight model with the old and new start of the drag rise, and Fluent (circles). Smooth wall in both.

### 4.3 Axial force along the flight

$$D_n(t) = q(t)\, A_{ref}\, C_{D,n}\big(M(t)\big) \tag{7}$$

with the friction of a 20 µm rough wall and the coupler included. It peaks at burnout: 278 N (A) and 234 N (B), 310 and 260 N on the hot day (Figure 7). Fluent, with a smooth wall, gives 282.6 and 237.6 N at the same point: the two agree within 2 %.

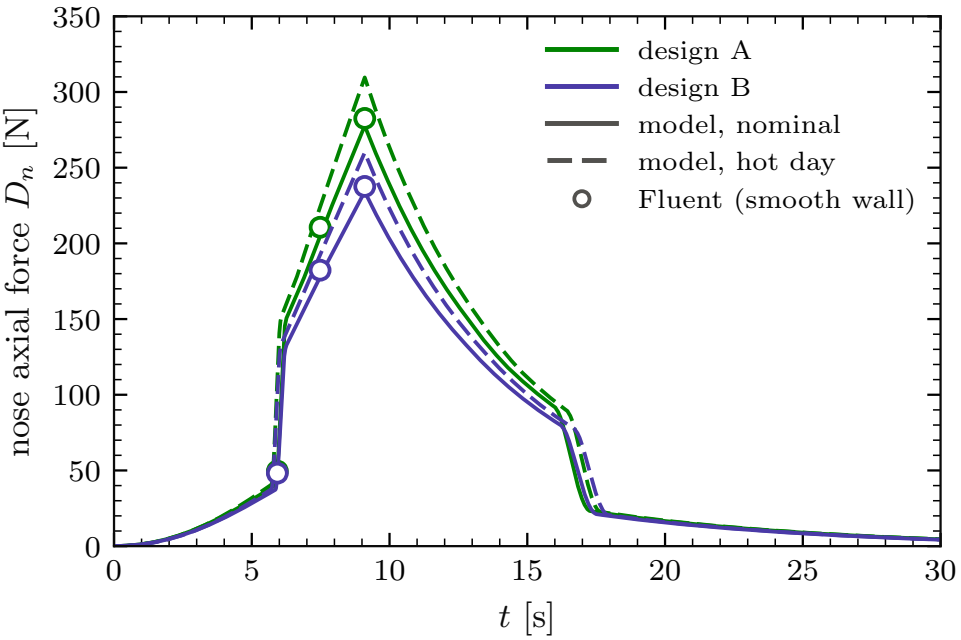



Figure 7: Aerodynamic axial force of the nose along the flight (model) and the Fluent points.

### 4.4 Normal force under incidence

Fluent was run at zero incidence only. Under a small incidence $\alpha$ [rad] slender-body theory gives the normal force of the nose and its distribution, with $S(x) = \pi r(x)^2$:

$$\frac{\mathrm{d}N}{\mathrm{d}x} = 2\,q\,\alpha\,\frac{\mathrm{d}S}{\mathrm{d}x}, \qquad N = 2\,q\,A_{ref}\,\alpha, \qquad x_{CP} = x_S - \frac{V_n}{A_{ref}}, \tag{8}$$

where $V_n$ is the volume of the nose. Per degree at LC1, $N/\alpha$ = 104 N (117 N hot day) for both designs; $x_{CP}$ = 295 mm (A) and 347 mm (B) from the tip. The load is carried mostly by the front half (Figure 8); the coupler cylinder carries none. The bending moment at the joint plane $x_j$ follows as $M_b = N\,(x_j - x_{CP})$; ST sets the design incidence (Req_12).

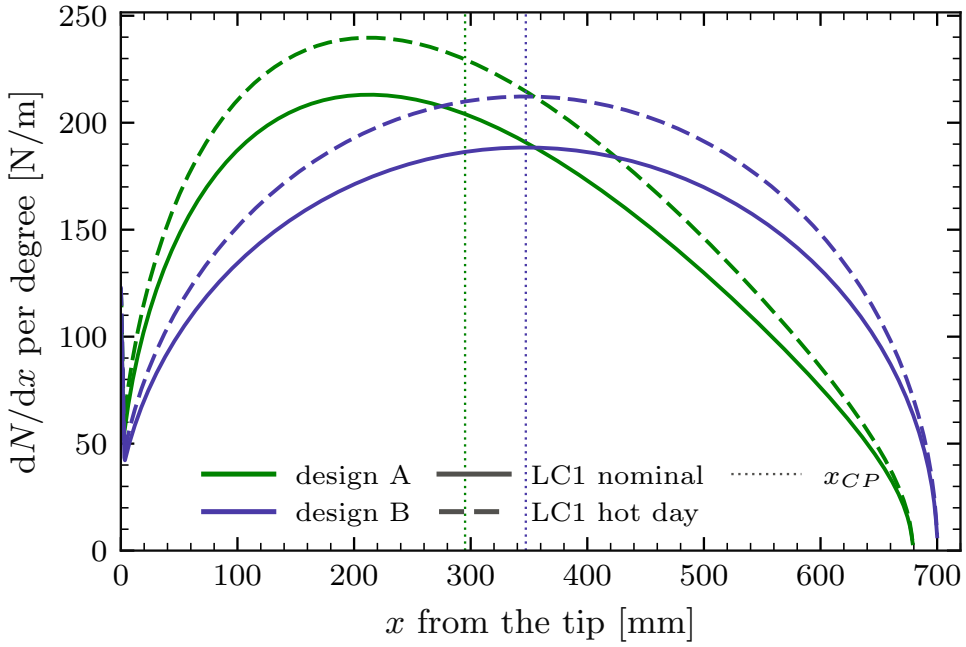



Figure 8: Normal load per unit length and per degree of incidence at LC1, slender-body theory; dotted: centre of pressure.

## 5 Heating

### 5.1 Fluent at burnout

Two runs per design at burnout give the heating along the wall:

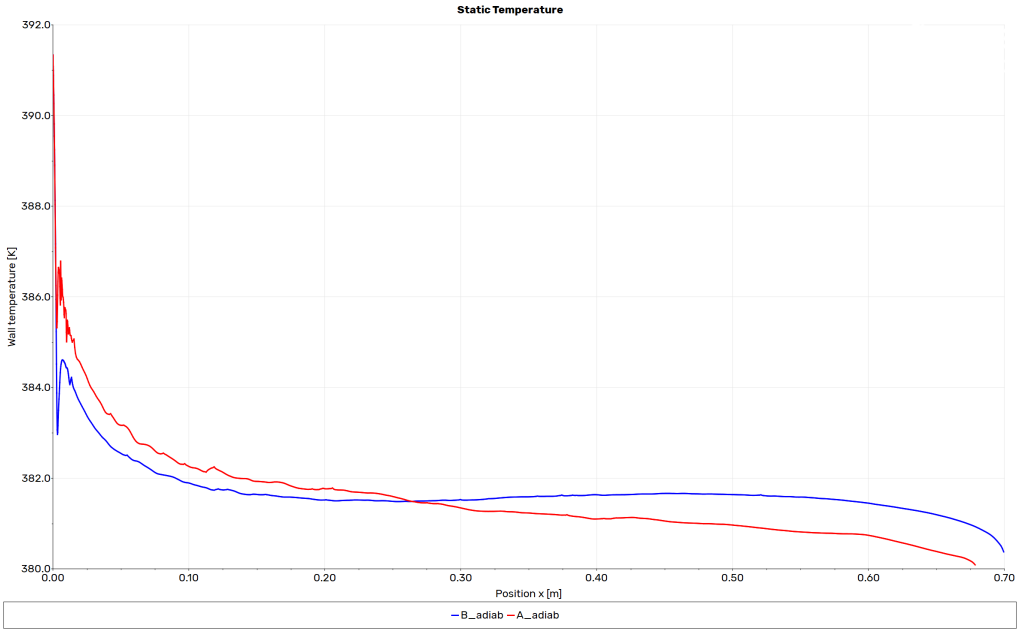
- **adiabatic wall** ($q_w = 0$): the wall takes the adiabatic wall temperature $T_{aw}(x)$, the temperature the air drives the wall towards;
- **wall at 300 K**: the heat flux $q_w(x)$ into a cold wall.

Together they give the heat-transfer coefficient and the recovery factor,

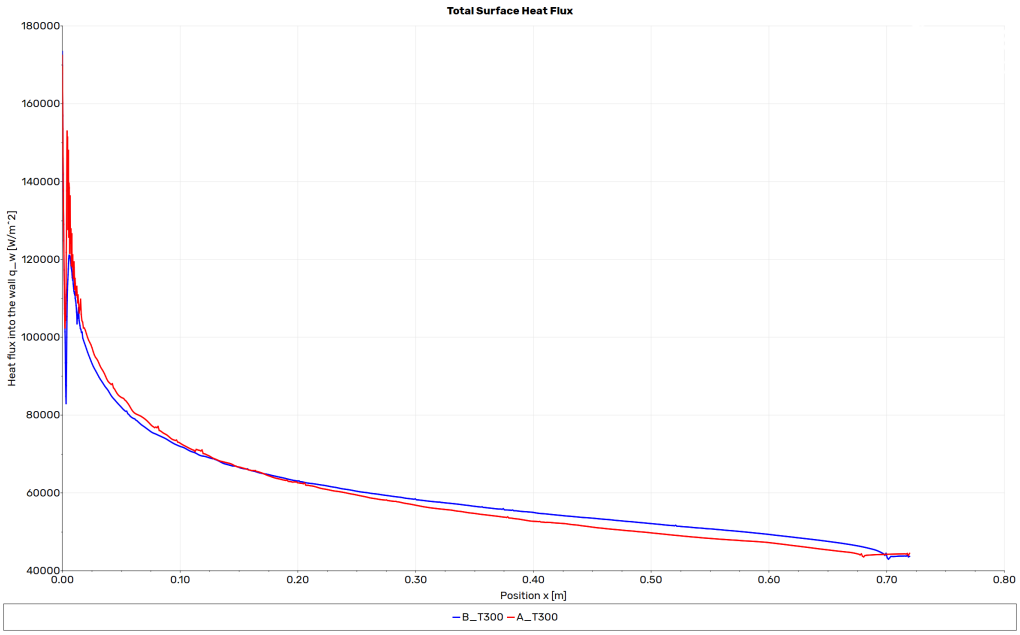
$$h(x) = \frac{q_w(x)}{T_{aw}(x) - 300\,\mathrm{K}}, \qquad r(x) = \frac{T_{aw}(x) - T_\infty}{T_0 - T_\infty}. \tag{9}$$

- **Tip**: $T_{aw}$ = 391.3 K, the stagnation temperature $T_0$ = 391.6 K. $q_w = 172.5$ (A) and 173.4 kW/m$^2$ (B); $h = 1888$ and 1899 W/(m$^2$K).
- **Shell**: $T_{aw}$ = 380–383 K beyond 50 mm, a recovery factor of 0.912 (A) and 0.914 (B). $h$ falls from about 1000 W/(m$^2$K) at 50 mm to 546 (A) and 539 W/(m$^2$K) (B) at the shoulder.

- The two designs differ only over the first 10 cm (Figure 9); the heating does not separate them.




(a) Adiabatic wall temperature $T_{aw}$.




(b) Heat flux into a wall at 300 K.

Figure 9: Heating along `wall_nose` at burnout, A (red) and B (blue), Fluent XY plots.

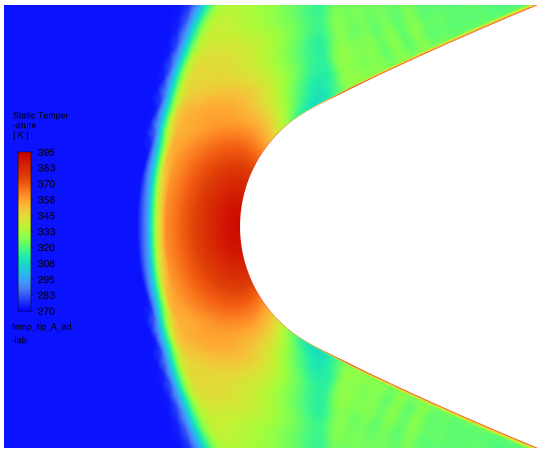



Figure 10: Air temperature around the tip of A at burnout, adiabatic wall (Fluent). The air is heated to $T_0$ in the subsonic region behind the bow shock and in the boundary layer.

## 5.2 Calibration of the flight model

Along the flight the heating comes from the Python model of report 1, which Fluent can now check:

$$h_s = 0.763\,\mathrm{Pr}^{-0.6}\sqrt{\rho_s\,\mu(T_0)\,\frac{\mathrm{d}u_e}{\mathrm{d}x}}\;c_p, \quad \frac{\mathrm{d}u_e}{\mathrm{d}x} = \frac{1}{r_n}\sqrt{\frac{2(p_0 - p_\infty)}{\rho_s}} \quad \text{(tip, Fay–Riddell)}, \tag{10}$$

$$h_t = \mathrm{St}\,\rho^* V c_p, \quad \mathrm{St} = 0.0296\,\mathrm{Re}_x^{*\,-0.2}\,\mathrm{Pr}^{-2/3}, \quad T_r = T_\infty\left(1 + r\,\tfrac{\gamma-1}{2}M^2\right),\ r = \mathrm{Pr}^{1/3} \quad \text{(shell, Eckert)}, \tag{11}$$

with $\rho^*$, $\mathrm{Re}_x^*$ at Eckert's reference temperature $T^* = T_\infty(1 + 0.032M^2 + 0.58(T_w/T_\infty - 1))$. At burnout, with the wall at 300 K:

Table 7: Fluent against the uncalibrated model at burnout, and the factors used in the flight model.

| Fluent / model | A | B | Factor |
|---|---|---|---|
| Heat flux at the tip | 0.943 | 0.949 | $k_s = 0.946$ |
| $h$ along the shell, 50 mm to shoulder (median, 10–90 %) | 1.010 (0.97–1.03) | 1.033 (1.00–1.04) | $k_t = 1.02$ |
| Heat into the first 80 mm (tip insert) [W] | 785 / 692 | 629 / 550 | $k_{tip} = 1.14$ |

The tip and shell correlations are within 6 % of Fluent. The lumped tip insert takes its lateral heating at $x_{tip}/2$, which misses the high $h$ just behind the spherical cap; with $k_s$ and $k_t$ applied it still gets 13–15 % too little heat, hence $k_{tip}$. Fluent's recovery factor (0.912) is a little above the model's $\mathrm{Pr}^{1/3} = 0.892$, so the model's driving temperature on the shell ($T_r = 378.9$ K) is 1–4 K below Fluent's $T_{aw}$.

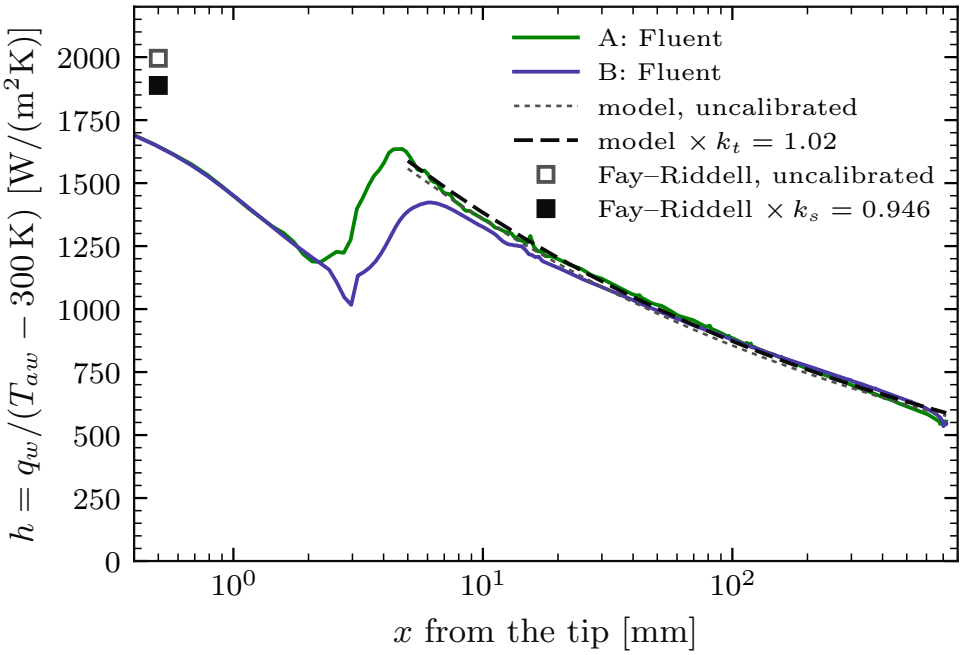



Figure 11: Heat-transfer coefficient at burnout on a 300 K wall: Fluent (smoothed between 3 and 15 mm), the shell correlation before and after $k_t$, and the tip value before and after $k_s$ (squares, placed at 0.5 mm).

### 5.3 Temperatures along the flight

With the calibrated $h$, the model integrates two wall models along the nominal and hot-day trajectories:

- **shell**: one-dimensional conduction through the CFRP, heated on the outside, adiabatic inside,

$$\rho c \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial y^2}, \qquad -k \frac{\partial T}{\partial y}\Big|_{out} = h\,(T_r - T_w), \qquad \frac{\partial T}{\partial y}\Big|_{in} = 0; \tag{12}$$

- **Al tip insert**: one temperature (aluminium conducts fast),

$$m\,c\,\frac{\mathrm{d}T}{\mathrm{d}t} = k_{tip}\left[0.55\,h_s A_{cap}(T_0 - T) + h_t(x_{tip}/2)\,A_{lat}(T_r - T)\right]. \tag{13}$$

Material data are those of report 1, to be confirmed by ST: CFRP 1.2 mm, $k = 0.7\,\mathrm{W/(m\,K)}$, $\rho = 1550\,\mathrm{kg/m^3}$, $c = 900\,\mathrm{J/(kg\,K)}$ (through the thickness); Al 6082 insert of 60 g over the first 80 mm, $c = 900\,\mathrm{J/(kg\,K)}$.

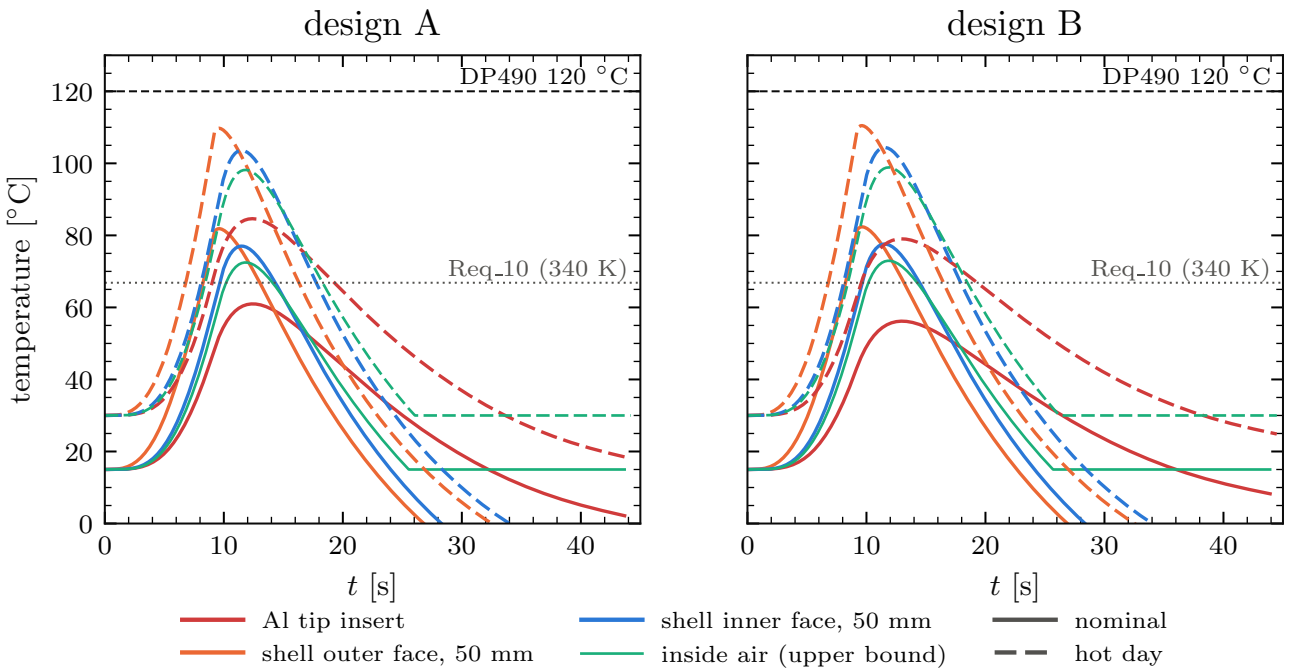



Figure 12: Temperatures along the flight, (a) A and (b) B: Al tip insert, CFRP shell at 50 mm (outer and inner face) and the upper bound of the inside air (mean inner-face temperature, not below $T_{pad}$).

The maxima are in Table 2. The outer face peaks just after burnout (9.6 s), the inner face 2 s later and the tip insert, with its larger heat capacity, at 12–13 s:

- the outer face of the shell stays below the DP490 limit of 120 °C, also on the hot day (110 °C);
- the inner face (77 °C) and the inside air (72 °C) exceed Req_10 (67 °C) on a nominal day; whether Req_10 applies to the air or to the wall is still open with SE;
- the tip insert of B gets 20 % less heat than A's (its front is thinner, so less surface in the first 80 mm) and stays 5–6 °C cooler.

For a thermal FEA, the data tables give $h$ and the driving temperature along the flight at the tip and at 20, 50, 80 and 150 mm, and $T_{aw}(x)$, $q_w(x)$, $h(x)$ at burnout (Appendix B).

## 6 Validity and limits

**Fluent runs.**

- **Convergence**: $(\max C_D - \min C_D)/\overline{C_D}$ of `wall_nose` over the last 500 iterations is 0.007–0.050 % in all eight runs, below the 0.1 % of REPORT-4.
- **Wall resolution**: largest $y^+$ 0.95 on the nose and 0.39 on the coupler, so the boundary layer is resolved down to the wall as SST requires.
- **Mesh**: the REPORT-4 recipe, 174 380 (A) and 174 155 (B) cells, maximum skewness 0.90 and 0.94. No mesh study was made for these two noses; the recipe was checked on study A only.
- **Tip sawtooth** in $C_p$, $q_w$ and $h$ between 3 and 15 mm, at the edge of the tip refinement. Smoothed in the delivered tables; a 40 mm refinement sphere would remove it. Integrated forces and the stagnation values are not affected.
- **Smooth wall**. The flight model uses a 20 µm roughness for the friction.

**From three runs to the whole flight.**

- **Zero incidence only** (2D axisymmetric). The normal force comes from slender-body theory, which is a first estimate for a nose alone; a 3D run at incidence would be needed to refine it.
- **Transonic rise** between $M = 0.95$ and 1.1 is not resolved. It affects the drag for about 1 s and none of the load cases.
- **Hot day** is not run in Fluent: the nominal $C_p(x, M)$ is applied with the hot-day $p_\infty$, $q$ and $M$. Above $M = 1.465$ $C_p$ is held, which puts the hot-day tip pressure 2 % low (Section 3).
- **Inside pressure** of the sealed nose is bounded, not computed: between the pad temperature and the mean inner-face temperature. Leaks or vents would bring it towards $p_\infty$ and reduce the outward loads.

**Heating.**

- Fluent gives the heating at burnout only, steady. Along the flight the calibrated correlations are used; they match Fluent at burnout within a few per cent but are not checked at other instants.
- Wall models: one-dimensional conduction through the shell (no conduction along it, adiabatic inside, no radiation), which is conservative for the shell peaks; and a single temperature for the tip insert, which is its mean temperature: the front of the insert runs hotter than the value given.
- Material data are assumptions of report 1 and must be confirmed by ST.

**Flight model.** The trajectory is the one of report 2 for the 720 mm noses, with the drag of the rest of the vehicle scaled so that the 650 mm baseline reaches 9 km. A change of the vehicle mass or motor changes $q$, and the pressures and forces scale with it.

# A CFD setup: differences from REPORT-4

The runs follow the study-A workflow of REPORT-4 (geometry, mesh, models, boundary conditions and solvers). Only these points differ.

- **Domain**: built in Ansys Discovery from the 3D STEP of each nose: the end disc is pulled to $x = 2520$ mm, a $6120 \times 3600$ mm rectangle on the XY plane of the origin is subtracted from it. Area 21.8233 m$^2$ (A) and 21.8269 m$^2$ (B), within 0.001 % of the domains of `CFD/make_domain_2d.py`.
- **Named selections**: the body line is split at the end of the coupler, $x = 720$ mm, into `wall_coupler` and `wall_body`; seven zones in all.
- **Mesh**: `wall_body` edge sizing hard, from 0.55 mm at $x = 720$ mm to 1.65 mm at the outlet; inflation scoped on the face with the three walls as boundary.
- **Fluent**: 2024 R2 started on its own and set up by script (`CFD/study720/setup_fluent.py`, `run_queue.py`, PyFluent). Turbulence intensity set to 1 % (the standalone default is 5 %). Report definitions `cd-total` (three walls) and `cd-nose` (`wall_nose`); residual stopping off.
- **Iterations**: 500 first order at Courant 1, 500 at Courant 5, then second order at Courant 5 up to 4000. $M = 0.95$: pressure-based coupled solver, 1200 iterations. 300 K walls: 3500 more iterations from the adiabatic solution.

Scripts, wall exports, $C_D$ histories, figures and meshes are in `CFD/study720/` of the repository (README there); the case and data files stay on the DLL PC.

# B Data tables for ST

Six CSV files in `report_720/data/`, one set per design (`A_`, `B_`), written by `report_720/figures.py`. $x$ is measured from the blunted tip along the axis; pressures are absolute; $\Delta p > 0$ pushes the shell inwards. Between $x = 3$ and 15 mm the Fluent values are smoothed over 2 mm.

Table 8: Contents of the data tables.

| File | Columns |
|---|---|
| `*_wall_x.csv` | One row per Fluent wall face (nose and coupler). `x_mm`, `r_mm`; `Cp_M095`, `Cp_M120`, `Cp_M147` (Fluent); for LC1–LC3, nominal and hot day, `p_wall_`*LC*`_`*case*`_kPa` and `dp_`*LC*`_`*case*`_kPa` ($p_{in}$ at $T_{pad}$ for LC1, LC2; heated air for LC3); at burnout `T_aw_LC1_K`, `q_w_300K_LC1_kW_m2`, `h_LC1_W_m2K`. |
| `*_history_t.csv` | One row per time step (0.02 s), `case` = nominal or hot. `t_s`, `M`, `q_Pa`, `p_inf_Pa`; `p_in_low_Pa` ($T_{pad}$) and `p_in_high_Pa` (heated air); at the tip, 50 mm, mid-length and shoulder `p_wall_`*st*`_Pa`, `dp_`*st*`_low_Pa`, `dp_`*st*`_high_Pa`; `D_n_N`; temperatures `T_tip_insert_C`, `T_shell_out_50mm_C`, `T_shell_in_50mm_C`, `T_air_upper_C`; thermal boundary conditions (calibrated, wall at 300 K) `h_stag_W_m2K` with `T0_K`, and `h_`*x*`mm_W_m2K` with `T_r_`*x*`mm_K` at 20, 50, 80 and 150 mm. |
| `*_normal_load_x.csv` | `x_mm`, `r_mm`, normal load per unit length and per degree at LC1, `dNdx_per_deg_nominal_N_per_m` and `dNdx_per_deg_hot_N_per_m`. |